

UET University Peshawar

Abbottabad Campus



Lab Report No: 03

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Class No: 21

Course: SE-305L Computer Communications and Networks

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Lab 3: Basic Router Configuration and Network Connectivity

1. Objectives: In this lab, you will;

- Configure a router and PCs in Cisco Packet Tracer.
- Set up basic switch connectivity.
- Verify network communication.
- Use `show` commands to check configurations.
- Save settings to ensure persistence.

2. Required Software & Tools:

- A computer with Windows OS
- Packet Tracer

3. Step by Step Instructions:

Step 1: Open Cisco Packet Tracer

Step 2: Add Network Devices

Step 3: Connect the Devices

Step 4: Assign IP Addresses to PCs

Step 5: Configure the Router

Step 6: Save the Configuration

Step 7: Test Connectivity

Step 8: Verify Configurations

Task 1 is consisting of step # 1,2 and 3 in the step-by-step instructions.

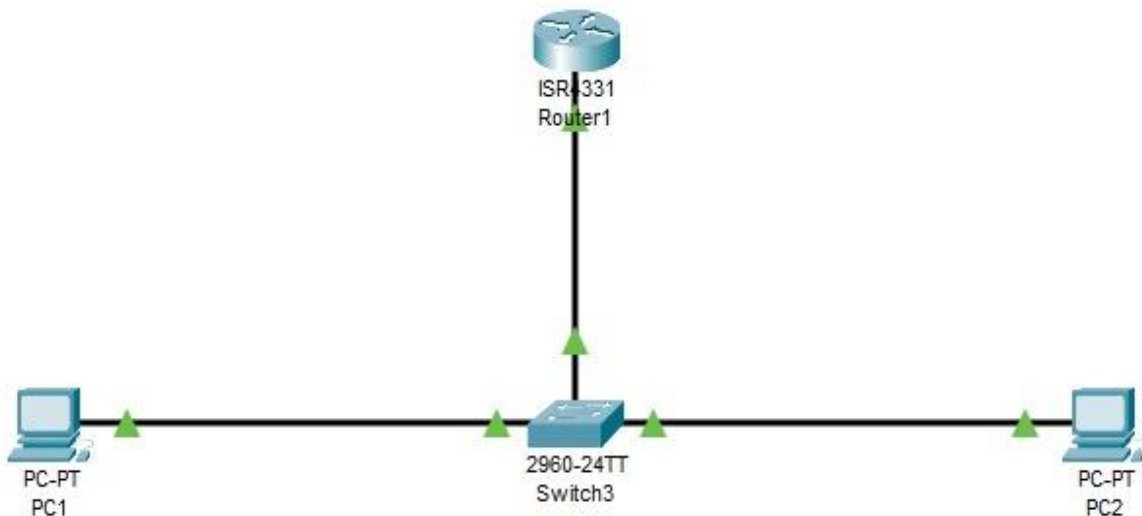


Figure 1 Network Diagram

Task 2 is consisting of step # 4,5 and 6 in the step-by-step connection.



The screenshot shows a Cisco Router CLI interface with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the IOS Command Line Interface. The user has entered several commands to configure the router, including setting the line protocol on GigabitEthernet0/0/0, copying the running configuration to the startup configuration, and displaying the current configuration. The output shows the current configuration, including the hostname MyRouter, version 16.6.4, and various service settings. The user has also entered commands to configure IP and IPv6 on the router interfaces.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
exit
MyRouter(config)#exit
MyRouter#
%SYS-5-CONFIG_I: Configured from console by console

MyRouter#copy running-config startup-config
Destination filename [startup-config]?
?Bad filename
%Error parsing filename (Bad file number)
MyRouter#
MyRouter#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
MyRouter#
?Bad filename
%Error parsing filename (Bad file number)
MyRouter#
?Bad filename
%Error parsing filename (Bad file number)
MyRouter#
MyRouter#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
MyRouter#show running-config
Building configuration...

Current configuration : 740 bytes
!
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname MyRouter
!
!
!
enable secret 5 $1$mERr$dc4d.y029TcPHc61hLGWV0
!
!
!
!
!
!
ip cef
no ipv6 cef

MyRouter#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
MyRouter#
```

Figure 2 IP's and imp configurations

Task # 3 is consisting of step # 7 and 8 respectively in the step-by-step connection.

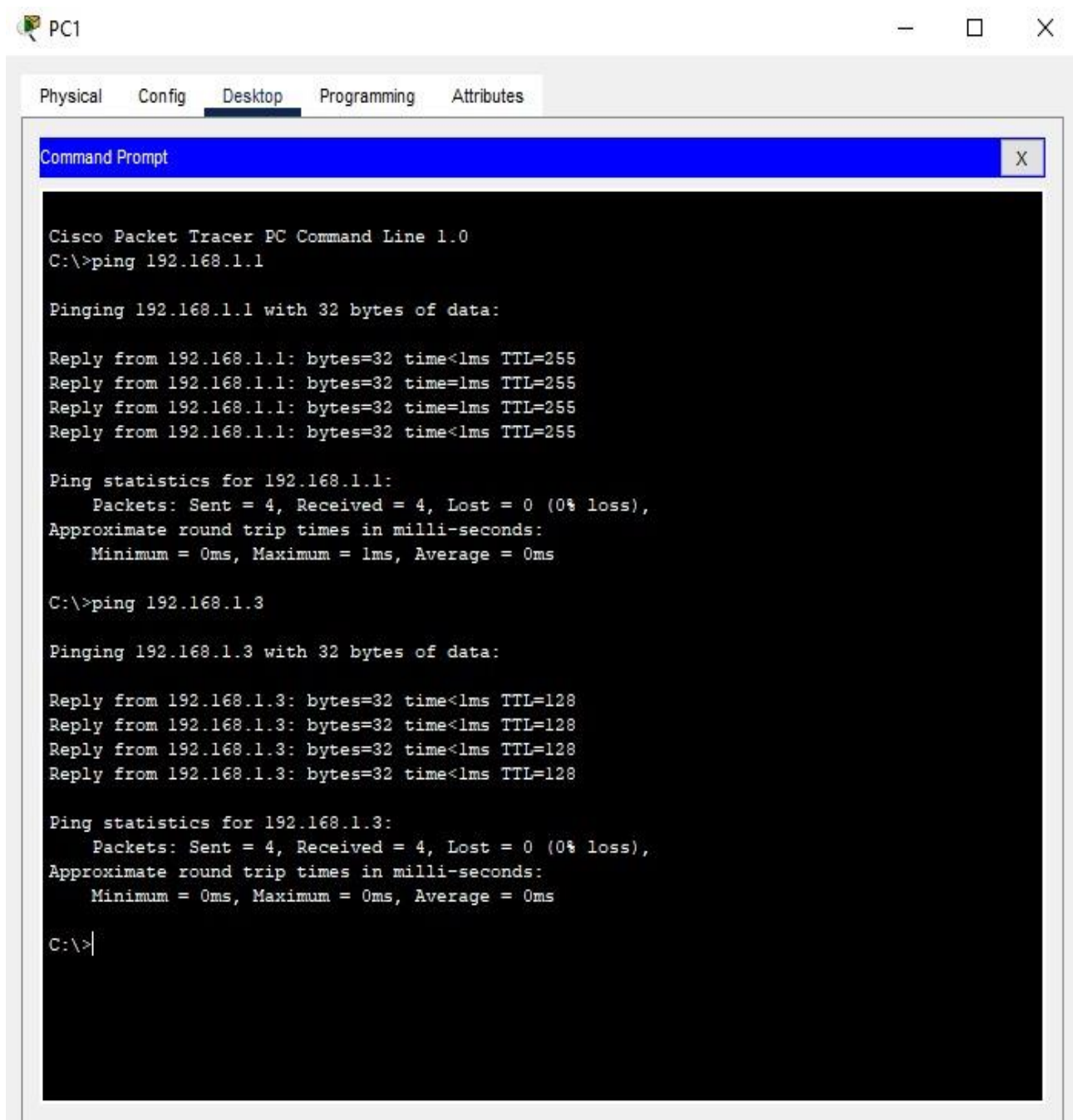


Figure 3 Testing Connectivity.

That's It!